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COMP3004- Team Project Use Cases

The following are the use cases developed based on the assignment specification:

USE CASE 1: AED DEFIBRILLATION OPERATION
Primary Actor: User.
Scope: AED Plus Device.
Level: User goal.
Goal in context: To perform actions comprising evaluating heart rhythms, delivering electric shocks, and presenting guidance during an emergency. Specifically, treat cardiac arrhythmia though AED defibrillation accomplished by delivering an electrical shock to the heart to restore a normal rhythm.
Stakeholders and Interests:
User – wants to use the AED Plus device to diagnose and treat patient during an emergency.
Patient – wants to receive necessary treatment in the event of a life-threatening cardiac emergency.
Precondition: AED Plus device is accessible, functioning properly and ready for use.
Minimal guarantees: AED Plus device self-test is successful.
Success guarantees: User can deliver an electrical shock to the patient's heart to restore a normal rhythm. Thus, the AED Plus device is operational and all phases are executed as intended.
Main success scenario:
1. The user retrieves the AED Plus device.
2. The user power on (Use Case 2).
3. The user electrode placement (Use Case 3).
4. The AED Plus device provides heart rhythm analysis (Use Case 4).

USE CASE 2: POWER ON
Primary Actor: User.
Scope: AED Plus Device.
Level: Subfunction.
Goal in context: Ensure the AED Plus device is powered and prepared for use.
Stakeholders and Interests:
User – wants to power on AED Plus device.
Patient – wants to receive necessary treatment in the event of a life-threatening cardiac emergency.
Precondition: AED Plus device is accessible.
Minimal guarantees: AED Plus device self-test is successful.
Success guarantees: AED Plus device displays visual as well as audible indications of being operational.
Main success scenario:
1. The user presses the "ON" button to turn on the AED Plus device.
2. The AED Plus device initiates a self-test to ensure that the device is functioning properly and ready for use.

3. The AED Plus device provides a visual and computer-generated audible indication that it is operational.
Extensions:
1a. The AED Plus device does not turn on as a result of there being no detectable electrical activity.
    1a1. The user replaces the batteries located in the back of the AED Plus device.
2a. The AED Plus device self-test is unsuccessful.
    2a1. The user manually performs test by holding the "ON" button for at least 5 seconds and changing the batteries if necessary.

USE CASE 3: ELECTRODE PLACEMENT
Primary Actor: User.
Scope: AED Plus Device.
Level: Subfunction.
Goal in context: Correctly position the electrode pads on the patient's chest.
Stakeholders and Interests:
User – wants to accurately place the adhesive electrode pads on the patient's bare chest.
Patient – wants to receive necessary treatment in the event of a life-threatening cardiac emergency.
Precondition: AED Plus device is powered on, functioning properly and ready for use.
Minimal guarantees: Electrode pads are correctly attached to the patient's chest.
Success guarantees: AED Plus device prompts that electrode placement is successful.
Main success scenario:
1. The AED Plus device prompts the user to place a set of adhesive electrode pads on the patient's bare chest.
2. The user retrieves the adhesive electrode pads that are included with the AED Plus device.
3. The user selects the electrode pads labelled "Adult Pads" if the patient is an adult or selects the electrode pads labelled "Child Pads" if the patient is a child.
4. The user places the set of adhesive electrode pads on the patient's bare chest as indicated in the AED prompts.
Extensions:
4a. The AED Plus device prompts the user to plug in the pads' connector to the AED.
    4a1. The user plugs in the pads' connector to the AED.

USE CASE 4: HEART RHYTHM ANALYSIS
Primary Actor: AED Plus device.
Scope: AED Plus Device.
Level: Subfunction.
Goal in context: Determine if a shock is necessary.
Stakeholders and Interests:
User – wants to provide necessary treatment based on AED diagnoses and prompts.

Patient – wants to receive necessary treatment in the event of a life-threatening cardiac emergency.
Precondition: Adhesive electrode pads are correctly placed on patient.
Minimal guarantees: AED will provide a heart rhythm analysis.
Success guarantees: AED accurately detects if a shock is necessary based on the patient's heart rhythm.
Main success scenario:
1. The AED Plus device provides a clear voice and visual prompt indicating to "Stand clear" implying to not make contact with the patient.
2. The AED Plus device provides a clear voice and visual prompt stating "Analyzing".
3. AED Plus device automatically analyzes the patient's heart / ECG rhythm through the electrodes.
4. AED Plus device monitors the electrical activity of the heart to determine whether a shockable rhythm is present; ventricular fibrillation and ventricular tachycardia.
Extensions:
1a. The user touches the patient during the heart rhythm analysis.
    1a1. The AED hearth rhythm analyses results may falsely indicate the required action.
4a. The AED Plus device provides a clear voice and visual prompt that a shock is necessary with the dialogue "Shock advised".
    4a1. The AED Plus device automatically charges itself.
    4a2. The AED Plus device tells the user when to press the button that will deliver the shock.
    4a3. The user provides shock delivery (Use Case 5).
    4a4. The AED Plus device instructs user to perform cardiopulmonary resuscitation (CPR) for a specified duration.
    4a5. The AED Plus device continues to monitor the patient's heart rhythm and provide feedback on the quality and rate of chest compressions during CPR and further shock or CPR instructions as needed.
4b. The AED Plus device provides a clear voice and visual prompt that a shock is not advised with the dialogue "Shock not advised".
    4b1. The AED Plus device instructs user to perform cardiopulmonary resuscitation (CPR) for a specified duration.
    4b2. The AED Plus device continues to monitor the patient's heart rhythm and provide feedback on the quality and rate of chest compressions during CPR.

USE CASE 5: SHOCK DELIVERY
Primary Actor: User.
Scope: AED Plus Device.
Level: Subfunction.
Goal in context: Correctly use AED Plus device to deliver a shock to the patient's heart.
Stakeholders and Interests:
User – wants to accurately deliver a shock to the patient's heart.

Patient – wants to receive necessary treatment in the event of a life-threatening cardiac emergency.
Precondition: AED Plus device advises and prompts user to deliver shock.
Minimal guarantees: AED Plus device charges itself in preparation for delivering the shock.
Success guarantees: The user accurately delivers a shock following the AED Plus device prompts.
Main success scenario:
1. The AED Plus device provides a clear voice and visual prompt indicating to “Stand clear” implying to not make contact with the patient.
2. The user presses the “Shock” button to administer the shock to the patient.
3. The user waits for further instructions from the AED Plus device.